License Plate Detection

ITAI 1378 - Computer Vision and AI

Midterm Project Proposal

Tier 1 - Core

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The Problem

Manually identifying license plates for over 10,000+ vehicle images daily is slow and inefficient taking an average 20 seconds per image.

Parking and traffic monitoring systems lose thousands of dollars every month in delayed entry and unbilled access.

Automating license plate detection delivers faster processing, consistent results, and scalable operations.



Our Solution

A computer vision application that automatically detects license plates using a YOLO object detection model.



01

Vehicle Image

YOLO

Object Detection Model

02

YOLO Object Detection Model



03

Detected License Plate (Annotated Output)

Computer Vision Technique

Object Detection with **YOLO**

- Single regression
- Fast inference speeds
- Low false-positive predictions

Ultralytics YOLO with **PyTorch**

Simple and easy plate detection for one-class dataset



Data Plan

License Plate Dataset by Ronak Gohil on Kaggle, 1,695 vehicle images. https://www.kaggle.com/datasets/ronakgohil/license-plate-dataset

Single class: license_plate; bounding-box annotations are in YOLO format.

Images are reviewed for coordinate accuracy and normalized to 640 x 640 resolution.

Reviewed for errors, then split into 80% training, 10% validation, and 10% test sets

A pre-annotated dataset minimizes data preparation and fits the Tier 1 scope timeline



Success Metrics

Detection accuracy

mAP@50 ≥ 0.70

measures bounding-box detection accuracy

Precision ≥ 0.80

measures correctness of detected license plates

Inference Latency (<= 100 ms per image)

measures the total time delay between input and output



Model reliably detects license plates across diverse vehicle images with few false positives

Milestone Plan

01	Submit the proposal, outline project architecture, and create the GitHub repository	Week 10
02	Run a pretrained YOLO model end to end on a few license-plate images	Week 11
03	Prepare and validate the Kaggle data, fine-tune YOLO, show training curves, build the crop/annotate pipeline	Week 12 & 13
04	Run the held-out test set and record mAP@50, precision, and latency; fix issues such as glare	Week 14
05	Demo video, final README, final slides	Week 15

Risks & Resources

01

Dataset Quality

inconsistent, missing, or hard-to-detect plates

→

Plan B

clean annotations, apply data augmentation, or switch to alternate public dataset

02

Model Performance

issues with lighting, angles, distances, occlusions

→

Plan B

further fine-tuning, adjust training parameters, limit scope to clear plate images

Compute & Cost

Google Colab free GPU resources
Kaggle free

Tools

Ultralytics YOLO +
PyTorch (open source)

Expected project cost

$0